

Business Intelligence definition

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1. Define Business Intelligence (BI).. Easy Answer

Business Intelligence (BI) is the process of collecting, analyzing, and presenting business data to help organizations make better decisions.

It uses tools and techniques to turn raw data into useful information like reports, dashboards, and charts for strategic planning and performance improvement.



2. What is the role of Metadata in a Data Warehouse?. Easy Answer

Metadata in a **Data Warehouse** is the "data about data."

It describes the structure, meaning, and source of the stored data — helping users understand where data comes from, how it is organized, and how to use it effectively.

👉 **In short:** Metadata acts as a guide or dictionary for the Data Warehouse.



3. Mention two differences between OLAP and OLTP.. Easy Answer

OLAP (Online Analytical Processing)	OLTP (Online Transaction Processing)
Used for analysis and reporting of data.	Used for day-to-day transaction processing.
Deals with large volumes of historical data .	Deals with current, real-time data .



4. Explain the relation between Business Intelligence (BI) and Data Warehousing (DW).. Easy Answer

Business Intelligence (BI) and **Data Warehousing (DW)** are closely related.

A **Data Warehouse** stores and organizes large amounts of data from different sources, while **Business Intelligence** uses that data to create reports, dashboards, and insights for decision-making.

👉 **In short:**

DW is the **data storage system**, and **BI** is the **analysis and reporting process** that uses that data.



5. List and briefly describe the defining features of a Data Warehouse.. Easy Answer

The main **defining features of a Data Warehouse** are:

1. 🧩 **Subject-Oriented** – Organized around key business subjects like sales, customers, or products.
2. ⌚ **Time-Variant** – Stores historical data for trend analysis over time.
3. 🚫 **Non-Volatile** – Data is stable; once entered, it is not frequently changed or deleted.
4. 🔗 **Integrated** – Combines data from multiple sources into a single, consistent format.



6. Differentiate between Data Warehouse and Data Mart.. Easy Answer

Data Warehouse	Data Mart
Stores data for the entire organization .	Stores data for a specific department or function .
Contains large volumes of data from many sources.	Contains a smaller, focused subset of data.



7. What are some common OLAP operations used in data analysis?. Easy Answer

Some common **OLAP operations** used in data analysis are:

1. 🔍 **Roll-up** – Summarizes data (e.g., from days to months).
2. 📉 **Drill-down** – Shows detailed data (e.g., from months to days).
3. 🔗 **Slice** – Selects a single dimension of data.
4. 🧩 **Dice** – Selects data from multiple dimensions.
5. ↔ **Pivot (Rotate)** – Changes the view or perspective of data for analysis.



8. Describe the lifecycle of data in BI.. Easy Answer

The **lifecycle of data in Business Intelligence (BI)** includes the following steps:

1. 📥 **Data Collection** – Gathering data from various sources.
2. 🧹 **Data Cleaning** – Removing errors and inconsistencies.
3. 🗄️ **Data Storage** – Storing data in a Data Warehouse.
4. 📊 **Data Analysis** – Using tools to analyze and find patterns.
5. 📈 **Data Visualization** – Presenting results using reports, charts, or dashboards.
6. 🧠 **Decision Making** – Using insights to make informed business decisions.



9. Explain the basic elements of Data Warehousing with examples.. Easy Answer

The **basic elements of Data Warehousing** are:

1.  **Data Sources** – Systems where data comes from (e.g., sales database, CRM).
2.  **ETL Process (Extract, Transform, Load)** – Extracts data from sources, cleans and converts it, then loads it into the warehouse.
3.  **Data Warehouse Database** – Central storage where all processed data is kept.
4.  **Metadata** – “Data about data” that describes the structure and meaning of stored data.
5.  **Data Access Tools** – Used for reporting and analysis (e.g., dashboards, BI tools like Power BI or Tableau).



10. Discuss trends in Data Warehousing.. Easy Answer

Some current **trends in Data Warehousing** are:

1.  **Cloud Data Warehousing** – Using cloud platforms like AWS, Google BigQuery, or Snowflake for flexible storage and scalability.
2.  **Automation and AI Integration** – Using AI to automate data cleaning and analysis.
3.  **Real-Time Data Processing** – Handling live data for instant insights.
4.  **Data Lake Integration** – Combining structured and unstructured data for deeper analysis.
5.  **Improved Data Security** – Using advanced encryption and access control to protect data.



11. Discuss in detail the building blocks of a Data Warehouse.. Easy Answer

The **building blocks of a Data Warehouse** are the core components that make it work effectively:

1.  **Data Sources** – These are the operational systems and external sources from which data is collected, such as ERP systems, CRM databases, or online transaction systems.
2.  **ETL (Extract, Transform, Load) Process** –
 - **Extract:** Collect data from different sources.
 - **Transform:** Clean, format, and integrate the data.
 - **Load:** Store the processed data into the Data Warehouse.
3.  **Data Storage** – The central repository where the cleaned and integrated data is stored. This includes:
 - **Fact Tables:** Store measurable, quantitative data (e.g., sales amount).
 - **Dimension Tables:** Store descriptive data for analysis (e.g., customer, product, time).
4.  **Metadata** – Provides information about the data, such as source, structure, meaning, and usage guidelines. It acts as a guide for users and tools.

5.  **Data Access Tools** – Tools that allow users to retrieve, analyze, and visualize data, such as dashboards, reports, OLAP tools, and BI platforms.
6.  **Management and Control** – Ensures proper operation, monitoring, and maintenance of the Data Warehouse, including security, performance, and data quality.

In short, these building blocks work together to collect, store, process, and present data for effective **decision-making**.



12. Explain dimensional analysis with a suitable example.
. Easy Answer

Dimensional Analysis in a Data Warehouse is a way of organizing data to make it easier to analyze using **facts** and **dimensions**.

- **Fact:** The measurable data (usually numeric).
- **Dimension:** The descriptive attributes that provide context to the fact.

Example:

A **sales fact table** might include:

- **Fact:** Sales Amount, Quantity Sold
- **Dimensions:** Time (day, month, year), Product (name, category), Customer (name, region)

This allows queries like:

- “What is the total sales amount of Product A in January 2025 in India?”

Here, **Time, Product, and Customer** are dimensions used to analyze the **Sales Amount** fact.



13. Compare and contrast OLAP and OLTP with examples.. Easy Answer

Feature	OLAP (Online Analytical Processing)	OLTP (Online Transaction Processing)
Purpose	Used for data analysis and decision-making .	Used for day-to-day transaction processing .
Data Type	Deals with historical, summarized data .	Deals with current, real-time data .
Operations	Complex queries like trends, forecasts, and reports .	Simple queries like insert, update, delete, retrieve .
Volume	Handles large volumes of data .	Handles smaller, routine transactions .
Example	Analyzing yearly sales trends across regions.	Recording a customer's purchase in an e-commerce system.



14. Describe the typical multitiered architecture of a data warehouse.. Easy Answer

The **typical multitiered architecture of a Data Warehouse** usually has **three tiers**:

1.  **Bottom Tier (Data Sources & ETL)**
 - Includes operational databases and external sources.
 - Data is extracted, cleaned, and transformed before loading into the warehouse.
2.  **Middle Tier (Data Warehouse Server / Storage)**
 - Central repository that stores integrated and historical data.
 - Can use **OLAP servers** for complex analysis.
3.  **Top Tier (Front-End / Presentation Layer)**
 - Tools for querying, reporting, and visualization.
 - Examples: dashboards, BI tools, and analytical reports.

In short: Bottom tier → Middle tier → Top tier = **Data Collection** → **Storage** → **Analysis & Reporting**.



15. Explain the differences between star schema, snowflake schema, and fact constellation schema advantages?. Easy Answer

Here's an **easy comparison** of the three common Data Warehouse schemas:

Schema	Structure	Advantages / Example
Star Schema	Fact table at the center connected to denormalized dimension tables .	Simple to understand, fast for queries. Example: Sales fact table with Product, Customer, and Time dimensions.
Snowflake Schema	Fact table connected to normalized dimension tables (dimensions may have sub-tables).	Saves storage, supports complex queries. Example: Customer dimension split into Customer → Address → Region.
Fact Constellation Schema	Multiple fact tables share dimension tables (also called Galaxy Schema).	Supports multiple processes, flexible for complex analysis. Example: Sales and Inventory fact tables sharing Product and Time dimensions.



16. Define Data Mining.. Easy Answer

Data Mining is the process of discovering **hidden patterns, trends, and useful information** from large sets of data using techniques like statistics, machine learning, and database systems.

 **In short:** It turns raw data into meaningful insights for decision-making.



17. What is the KDD process?. Easy Answer

The **KDD (Knowledge Discovery in Databases) process** is a series of steps to find useful knowledge from data:

1.  **Data Selection** – Choose relevant data from databases.
2.  **Data Preprocessing** – Clean and prepare the data.
3.  **Data Transformation** – Convert data into suitable formats for mining.
4.  **Data Mining** – Apply algorithms to discover patterns.
5.  **Pattern Evaluation** – Identify useful and meaningful patterns.
6.  **Knowledge Presentation** – Present results through reports, charts, or dashboards.



18. List two data reduction techniques.. Easy Answer

Two common **data reduction techniques** are:

1. **Dimensionality Reduction** – Reducing the number of variables or features in the data (e.g., using PCA).
2. **Data Aggregation** – Summarizing data to a higher level (e.g., total sales per month instead of per day).



19. Explain the functionalities of Data Mining.. Easy Answer

The **functionalities of Data Mining** include:

1.  **Classification** – Assigning data to predefined categories (e.g., classifying emails as spam or not).
2.  **Regression** – Predicting numeric values (e.g., forecasting sales).
3.  **Clustering** – Grouping similar data together (e.g., customer segmentation).
4.  **Association Rule Mining** – Finding relationships between items (e.g., “customers who buy bread also buy butter”).
5.  **Trend and Sequence Analysis** – Detecting patterns over time (e.g., stock market trends).



20. Differentiate between data cleaning and data integration.. Easy Answer

Feature	Data Cleaning	Data Integration
Purpose	Removes errors, inconsistencies, and missing values from data.	Combines data from multiple sources into a unified view.
Focus	Quality and accuracy of data.	Consolidation of data for analysis.
Example	Correcting misspelled customer names.	Merging sales data from different regional databases.



21. What are data mining task primitives? Give examples.. Easy Answer

Data Mining Task Primitives are basic instructions that define **what kind of data mining task** to perform and **how to perform it**.

Examples:

1. **Task Type** – Specifies the goal: classification, clustering, association, or regression.
 - Example: Classifying customers as “High” or “Low” risk.
2. **Data Specification** – Defines the dataset or attributes to be used.
 - Example: Use **Customer Age, Income, Purchase History**.
3. **Interestingness Measures** – Criteria to evaluate patterns.
 - Example: Confidence $\geq 70\%$ for association rules.
4. **Background Knowledge** – Any prior information to guide mining.
 - Example: Known relationships between products.
5. **Presentation** – How the results are displayed.
 - Example: Charts, reports, or dashboards.



22. Explain various issues in Data Mining.. Easy Answer

Some common **issues in Data Mining** are:

1. ♦ **Data Quality** – Incomplete, noisy, or inconsistent data can affect results.
2. ♦ **Scalability** – Mining very large datasets can be slow and require high computing resources.
3. ♦ **Privacy and Security** – Sensitive data must be protected during mining.
4. ♦ **Integration of Data** – Combining data from multiple sources can be challenging.
5. ♦ **Choice of Algorithm** – Selecting the right data mining technique for the problem.
6. ♦ **Interpretation of Results** – Ensuring patterns discovered are meaningful and actionable.



23. Discuss the role of discretization and concept hierarchy in DM.. Easy Answer

In **Data Mining (DM)**, **discretization** and **concept hierarchy** help simplify and organize data for analysis:

1. **Discretization** – Converts continuous data into discrete intervals or categories.
 - Example: Converting **Age** from numbers to categories like **Young (0-25)**, **Adult (26-50)**, **Senior (51+)**.
 - **Role:** Makes data easier to analyze and discover patterns.
2. **Concept Hierarchy** – Organizes data attributes into a hierarchy from detailed to general.
 - Example: **City** → **State** → **Country** or **Product** → **Category** → **Department**.
 - **Role:** Helps in summarization, roll-up, and drill-down analysis in data mining.



24. Describe dimensionality reduction techniques with examples.. Easy Answer

Dimensionality Reduction is the process of reducing the number of attributes (features) in a dataset while retaining important information.

Techniques and Examples:

1. **Principal Component Analysis (PCA)** – Transforms data into a smaller set of uncorrelated variables.
 - Example: Reducing 10 financial indicators to 2-3 main components for analysis.
2. **Feature Selection** – Selecting only the most relevant attributes.
 - Example: Using only **Age**, **Income**, and **Purchase History** for predicting customer churn.
3. **Discretization / Binning** – Converting continuous attributes into intervals.
 - Example: Converting **Salary** into **Low**, **Medium**, **High** categories.

Role: Reduces computation, simplifies analysis, and improves model performance.



25. List the major tasks involved in data preprocessing. Explain the significance of each? Easy Answer

The **major tasks in Data Preprocessing** are:

1. 🗑️ **Data Cleaning** – Removes errors, missing values, and inconsistencies.
 - **Significance:** Ensures accurate and reliable analysis.
2. 🔗 **Data Integration** – Combines data from multiple sources into a unified dataset.
 - **Significance:** Provides a complete view for analysis.

3.  **Data Transformation** – Normalizes, aggregates, or converts data into suitable formats.
 - **Significance:** Makes data compatible for mining algorithms.
4.  **Data Reduction** – Reduces data volume using techniques like sampling or dimensionality reduction.
 - **Significance:** Improves processing speed and efficiency.
5.  **Data Discretization** – Converts continuous data into categorical intervals.
 - **Significance:** Simplifies analysis and pattern discovery.



26. What is Histogram ? Explain with example.. Easy Answer

A **Histogram** is a graphical representation of data that shows the **frequency distribution** of continuous or discrete values using bars.

- **X-axis:** Data intervals (bins)
- **Y-axis:** Frequency (number of occurrences)

Example:

Suppose the ages of 10 students are: 15, 16, 15, 17, 16, 18, 17, 16, 15, 18.

Age	Frequency
15	3
16	3
17	2
18	2

A histogram would show 4 bars representing ages 15–18 with heights corresponding to their frequencies.

Role: Helps to visualize data distribution and detect patterns or outliers.



27. What is Sampling ? Explain with example.. Easy Answer

Sampling is the process of selecting a **small, representative subset** of data from a larger dataset for analysis.

Example:

If a company has **10,000 customers**, instead of analyzing all 10,000, it can **sample 1,000 customers** randomly to study buying behavior.

Role:

- Reduces computation time.
- Makes data analysis faster and more efficient while still reflecting the overall dataset.



28. Explain the integration of a Data Mining system with a database or a Data Warehouse.. Easy Answer

The **integration of a Data Mining (DM) system** with a **Database or Data Warehouse** allows mining tools to access, analyze, and extract useful knowledge from stored data.

How it works:

1. **Data Access** – DM system connects to the database or warehouse to retrieve data.
2. **Data Preparation** – Cleans, transforms, and selects relevant data for mining.
3. **Mining Process** – Applies algorithms (like classification, clustering, or association) on the prepared data.
4. **Result Storage** – Stores discovered patterns or insights back in the database or presents them via reports.

Example:

A retail store's Data Warehouse contains sales and customer data. The DM system mines this data to find patterns like "Customers who buy bread often buy butter."

Role: Ensures efficient use of existing data for **decision-making and business intelligence**.



29. Discuss in detail the phases of the KDD process.. Easy Answer

The **KDD (Knowledge Discovery in Databases) process** has several key phases:

1.  **Data Selection** – Choose relevant data from different sources for analysis.
 - **Example:** Selecting sales and customer data from a company database.
2.  **Data Preprocessing** – Clean the data by handling missing values, noise, and inconsistencies.
 - **Example:** Correcting misspelled customer names or removing duplicate records.
3.  **Data Transformation** – Convert data into suitable formats or aggregate it for mining.
 - **Example:** Converting dates into months or categorizing ages into groups.
4.  **Data Mining** – Apply algorithms to discover hidden patterns and relationships.
 - **Example:** Using clustering to segment customers based on buying behavior.
5.  **Pattern Evaluation** – Identify useful and meaningful patterns from the mined data.
 - **Example:** Selecting association rules with high confidence, like "If a customer buys bread, they buy butter 80% of the time."
6.  **Knowledge Presentation** – Present the discovered knowledge in an understandable format.
 - **Example:** Using dashboards, charts, or reports for decision-making.

In short: Data is **selected** → **cleaned** → **transformed** → **mined** → **evaluated** → **presented** for actionable insights.



30. Explain data preprocessing steps in detail with examples.. Easy Answer

Data Preprocessing is the process of preparing raw data for analysis in Data Mining. The main steps are:

1. 🗑️ **Data Cleaning** – Remove errors, inconsistencies, and missing values.
 - **Example:** Correct misspelled customer names or fill missing ages with the average value.
2. 🔗 **Data Integration** – Combine data from multiple sources into a single dataset.
 - **Example:** Merging sales data from different regional databases into one table.
3. 🔄 **Data Transformation** – Convert data into a suitable format for analysis, including normalization, aggregation, or encoding.
 - **Example:** Converting salaries into a standard currency or scaling values between 0 and 1.
4. 📉 **Data Reduction** – Reduce data size using techniques like sampling, dimensionality reduction, or aggregation.
 - **Example:** Using only key attributes like Age, Income, and Purchase History instead of all 50 customer attributes.
5. 📁 **Data Discretization** – Convert continuous data into intervals or categories.
 - **Example:** Converting Age into groups like **Young** (0–25), **Adult** (26–50), **Senior** (51+).

Significance: These steps improve **data quality**, **reduce computation time**, and **make patterns easier to discover**.



31. What is attribute relevance?. Easy Answer

Attribute Relevance refers to how important or useful a specific attribute (feature) of the data is for a particular data mining task.

- **Example:** In predicting whether a customer will buy a product:
 - **Income** and **Age** might be **highly relevant**.
 - **Favorite Color** might be **less relevant**.

Role: Helps in **feature selection**, reducing irrelevant data and improving model accuracy.



32. Define support and confidence in Association Rule Mining.. Easy Answer

In Association Rule Mining:

1. **Support** – Indicates how frequently an itemset appears in the dataset.
 - **Example:** If 30 out of 100 transactions include both bread and butter, the support = 30%.
2. **Confidence** – Measures how often the rule is true, i.e., the likelihood that item B is bought when item A is bought.
 - **Example:** If 30 transactions include both bread and butter, and 40 transactions include bread, confidence = $30 \div 40 = 75\%$.

Role: Support and confidence help identify **strong and meaningful rules** from data.



33. Mention one difference between classification and association.. Easy Answer

Classification	Association
Predicts a target class for data (e.g., whether a customer will buy or not).	Finds relationships or patterns between items (e.g., customers who buy bread also buy butter).



34. Explain Market Basket Analysis.. Easy Answer

Market Basket Analysis (MBA) is a data mining technique used to find **relationships between items that are frequently bought together**.

- **Example:** A supermarket finds that customers who buy **bread** often also buy **butter**.
- **Role:** Helps in **product placement, cross-selling, promotions, and inventory management**.

It uses **Association Rule Mining** concepts like **support** and **confidence** to identify strong item combinations.



35. What is the role of concept description in Data Mining?. Easy Answer

Concept Description in Data Mining summarizes and describes the characteristics of a particular class or concept in the data.

- **Example:** For the concept “High-Value Customers,” concept description might show that they are usually **aged 30–50, live in urban areas, and spend more than \$500 per month**.

Role: Helps to **understand patterns, summarize data, and support decision-making**.



36. Differentiate between Apriori and Improved Apriori algorithms.. Easy Answer

Feature	Apriori Algorithm	Improved Apriori Algorithm
Candidate Generation	Generates all possible candidate itemsets.	Reduces the number of candidate itemsets using optimizations.
Performance	Slower for large datasets due to multiple database scans.	Faster and more efficient for large datasets.
Memory Usage	Higher memory usage.	Lower memory usage due to pruning techniques.



37. Explain the process of finding frequent itemsets using Apriori algorithm.. Easy Answer

The **Apriori Algorithm** finds frequent itemsets in a dataset using the **support threshold**. The process is:

- Generate 1-itemsets (C1)** – Count the frequency of each single item.
 - Keep items whose frequency $\geq$ minimum support $\rightarrow$ **L1** (frequent 1-itemsets).
- Generate 2-itemsets (C2)** – Combine L1 items to form pairs and count their frequency.
 - Keep itemsets with frequency $\geq$ minimum support $\rightarrow$ **L2**.
- Repeat** – Continue combining frequent (k-1)-itemsets to form k-itemsets (Ck) and prune infrequent ones $\rightarrow$ **Lk**, until no more frequent itemsets are found.
- Output** – All frequent itemsets (L1, L2, L3...) are used to generate association rules.

Example:

If transactions are {A, B}, {A, C}, {A, B, C} and minimum support = 50%:

- L1 = {A}, {B}, {C}
- L2 = {A, B}, {A, C}
- L3 = {A, B, C} if support $\geq$ 50%

This way, the algorithm identifies **frequent combinations of items**.



38. Describe associative classification with an example.. Easy Answer

Associative Classification is a data mining technique that combines **association rule mining** and **classification**.

- Process:**
 - Generate association rules from the dataset.
 - Select rules that predict the **class label**.

3. Build a classifier using these rules.

Example:

Dataset of customers: {Age, Income, BuysProduct}

- Rule: Age = 30–40 AND Income = High → BuysProduct = Yes
- This rule can classify new customers as likely to buy or not buy the product.

Role: Helps create **accurate classifiers** using patterns found in data.



39. Discuss class comparisons in concept description.. Easy Answer

Class Comparison in **Concept Description** compares different classes in a dataset to understand their distinguishing characteristics.

- **Purpose:** To find what makes one class different from another.

Example:

Dataset of customers with classes **High-Value** and **Low-Value**.

Attribute	High-Value	Low-Value
Age	30–50	18–30
Income	High	Low
Location	Urban	Rural

Role: Helps in identifying **key differences between classes** for decision-making and targeted strategies.



40. Explain in detail the Apriori algorithm with an example. Easy Answer

The **Apriori Algorithm** is used in **Association Rule Mining** to find **frequent itemsets** in a dataset. It works based on the principle that **all subsets of a frequent itemset must also be frequent**.

Steps of the Apriori Algorithm:

- 1. Generate Candidate 1-itemsets (C1):**
Count the frequency of each individual item in all transactions.
- 2. Find Frequent 1-itemsets (L1):**
Keep only the items whose frequency $\geq$ minimum support.
- 3. Generate Candidate 2-itemsets (C2):**
Combine frequent 1-itemsets (L1) to form pairs.
- 4. Find Frequent 2-itemsets (L2):**
Keep only pairs whose frequency $\geq$ minimum support.

5. Repeat:

- Generate k-itemsets from $L(k-1)$
- Count support and prune infrequent itemsets
- Continue until no more frequent itemsets are found.

6. Generate Association Rules:

Use the frequent itemsets to create rules and calculate **confidence**.

Example:

Transactions:

1. {A, B, C}
2. {A, B}
3. {A, C}
4. {B, C}

Minimum Support: 50% (2 out of 4 transactions)

Step 1: C1 → Count items

- A: 3, B: 3, C: 3 → $L1 = \{A, B, C\}$

Step 2: C2 → Generate pairs

- {A, B}, {A, C}, {B, C}

Step 3: L2 → Check support

- {A, B}: 2, {A, C}: 2, {B, C}: 2 → all ≥ 2 → $L2 = \{A, B\}, \{A, C\}, \{B, C\}$

Step 4: C3 → Generate triplet

- {A, B, C} → Support = 1 → less than 2 → discard

Result: Frequent itemsets = L1 and L2

Step 5: Generate rules (with confidence $\geq$ threshold)

- Rule: A → B with confidence = $2/3 = 66\%$

Role: Apriori efficiently finds **frequent item combinations** that help in **market basket analysis and decision-making**.



41. Discuss incremental association rule mining and its advantages.. Easy Answer

Incremental Association Rule Mining is a technique that **updates existing association rules** when new data is added, without reprocessing the entire dataset.

How it works:

1. Start with existing frequent itemsets and rules.
2. When new transactions arrive, update the support counts of existing itemsets.
3. Identify new frequent itemsets and generate new rules if needed.

Advantages:

1. 🕒 **Faster than full re-mining** – Only processes new data instead of the entire database.
2. 💾 **Efficient use of resources** – Reduces computation and memory usage.
3. 🔄 **Keeps rules up-to-date** – Handles dynamic datasets where transactions are continuously added.

Example:

A supermarket has rules based on last month's sales. When new sales data comes in this month, incremental mining updates rules like **Bread** → **Butter** without re-analyzing all previous transactions.



42. Write a note on rule generation and evaluation in ARM.. Easy Answer

In **Association Rule Mining (ARM)**, **rule generation and evaluation** are key steps:

1. Rule Generation:

- From **frequent itemsets**, generate **association rules** of the form $X \rightarrow Y$ (if X occurs, Y is likely to occur).
- Example: From {**Bread**, **Butter**} frequent itemset, generate rule **Bread** → **Butter**.

2. Rule Evaluation:

- Assess the **strength and usefulness** of the rules using:
 1. **Support:** How often the itemset appears in transactions.
 - Example: Bread and Butter appear together in 30 out of 100 transactions → Support = 30%
 2. **Confidence:** Likelihood that Y occurs when X occurs.
 - Example: If Bread appears in 40 transactions, Confidence = $30 \div 40 = 75\%$
 3. **Lift (optional):** Measures how much more likely Y occurs with X than by chance.

Role: Ensures that only **strong and meaningful rules** are considered for decision-making.